# Propulsion Architecture Trade Study for a Small VTOL ISR UAV

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## 1. Project Goal

The goal of this project is to perform a **system-level propulsion architecture trade study** for a small VTOL fixed-wing ISR UAV. Two representative VTOL propulsion concepts : **quad-lift + pusher** and **tilt-rotor**, are compared using physics-based performance estimates, mass and power considerations, and qualitative assessments of system complexity. The objective is to identify the key architectural trade-offs that influence hover efficiency, cruise performance, and overall system suitability for an ISR mission, without building or testing hardware.

## 2. Mission Definition

The defined mission represents a realistic tactical ISR operation for a small VTOL fixed-wing UAV in a defense and security context. The UAV supports ground forces operating in rural or semi-complex terrain with limited infrastructure. The primary role of the system is to provide real-time ISR (aerial intelligence, surveillance, and reconnaissance) to improve situational awareness, support maneuver planning, and enhance force protection.

The UAV is deployed by a small ground team from an unprepared site such as a field, roadside clearing, or forward operating base. Vertical takeoff and landing capability is required due to the absence of runways and the need for rapid deployment. After launch, the aircraft transits to a designated area of interest, where it performs persistent ISR operations, including area observation, target tracking, and reconnaissance. The mission concludes with a vertical recovery at or near the launch site.

The system is intended for day and night operations and must be capable of operating in moderately contested environments, including reduced GNSS availability and challenging weather conditions.

## 2.1 Mission Requirements

**Takeoff and Landing :**
The UAV shall be capable of vertical takeoff and landing from unprepared terrain without the use of a runway. Launch and recovery must be possible within confined areas and with minimal ground support equipment.

**Endurance and Range :**
The system shall provide a minimum total flight endurance of 90 minutes, including transit, on-station ISR, and return. The UAV shall maintain a reliable command, control, and data link at ranges of at least 15 kilometers in line-of-sight conditions.

**Altitude and Environment :**
The UAV shall be capable of operating up to an altitude of at least 3,500 meters above mean sea level. It shall support both daytime and nighttime missions and remain operable in moderate wind conditions.

**Payload :**
The UAV will carry an electro-optical and infrared (EO/IR) stabilized gimbal capable of providing full-motion video for surveillance and reconnaissance. The payload shall support real-time video transmission and basic onboard processing. The system shall allow for limited payload modularity for future sensor integration. (Figure 4).

**Data and Mission Operation :**
The UAV shall transmit real-time sensor data and telemetry to a ground control station. Mission execution shall support waypoint navigation, loitering patterns, and autonomous return and landing functionality.

**Mission Phases :**
The mission consists of vertical takeoff, climb to operational altitude, transit to the area of interest, persistent ISR on station, return flight, and vertical landing.

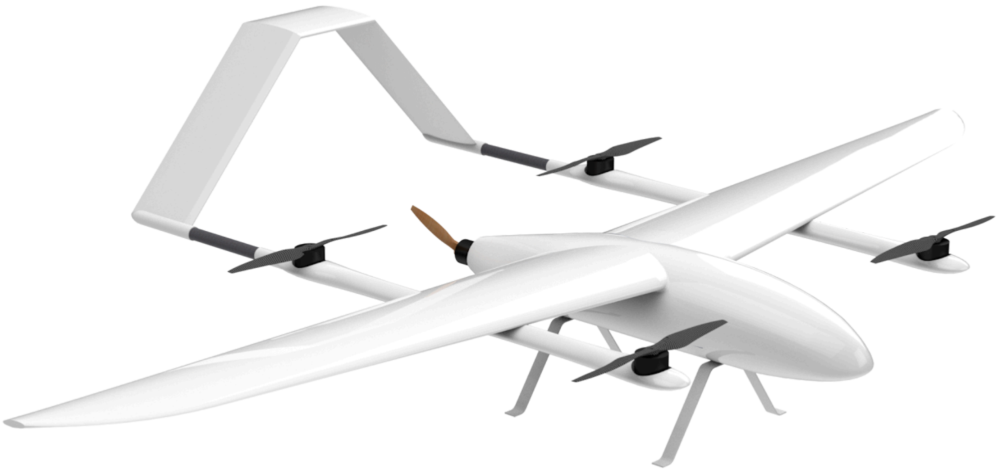
### 2.2 Performance Targets

| | |
|---|---|
| **MTOW:** | Target range between 3.5 and 7 kilograms. |
| **Endurance:** | Min 90 minutes total mission endurance. |
| **Operational Range:** | At least 15 kilometers line-of-sight communication range. |
| **Operating Altitude:** | Up to 3,500 meters above mean sea level. |
| **Payload Capability:** | Stabilized EO/IR sensor payload with real-time data transmission. |
| **Autonomy Level:** | Automated mission execution including waypoint flight, loitering, and autonomous recovery. |

## 3. Architectures Compared

This study compares two representative VTOL propulsion architectures commonly used in small fixed-wing ISR UAVs: quad-lift + pusher and tilt-rotor configurations. Representative examples of each architecture are shown in Figures 5–6 to illustrate the general layout and propulsion concept.

## 3.1 Quad-Lift + Pusher Architecture



The quad-lift + pusher configuration employs four vertically oriented lift rotors for takeoff, landing, and hover, combined with a dedicated pusher propeller for forward flight (Figure 5). During cruise, the lift rotors are inactive while the fixed wing provides lift. This architecture is widely used due to its mechanical simplicity, clear separation of flight modes, and robust VTOL control characteristics, at the expense of carrying inactive lift propulsion mass during cruise.

## 3.2 Tilt-Rotor Architecture

In the tilt-rotor configuration, the same rotors are used for both vertical lift and forward propulsion by mechanically tilting the thrust direction (Figures 6). This enables propulsion hardware reuse across all flight phases, reducing inactive mass during cruise. However, the architecture introduces additional mechanical complexity and tighter coupling between propulsion and flight control.

# 4. Block Diagrams

To enable a system-level comparison of the investigated VTOL UAV propulsion architectures, functional block diagrams were developed for each configuration. These diagrams represent the interaction between major subsystems in terms of **power flow, control, and data exchange**, without implying physical layout or detailed implementation.

The block diagrams form the architectural basis for the subsequent hover power, cruise performance, and mass trade analyses.

## 4.1 General Block Diagram Structure

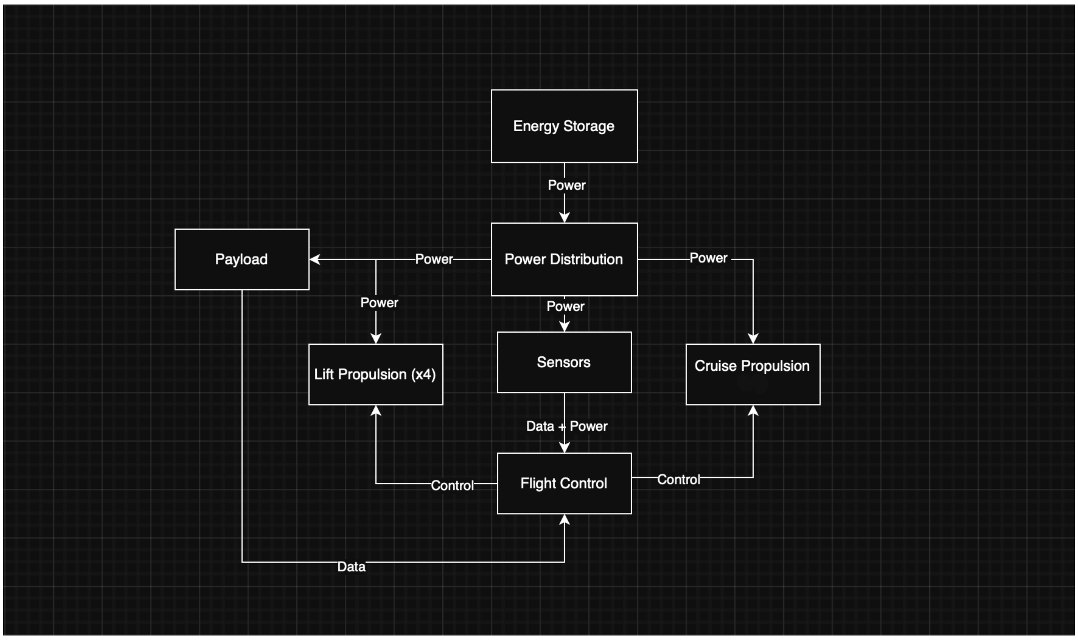
Both architectures share a common set of subsystems:

- Energy Storage
- Power Distribution
- Flight Control
- Sensors
- Payload (Gimbal, Camera (ISR))

## 4.2 Quad-Lift + Pusher Architecture

In the quad-lift + pusher configuration, propulsion is separated into two subsystems: **Lift Propulsion**, used for vertical flight, and **Cruise Propulsion**, used during forward flight. Each receives power from the Power Distribution block and control commands from Flight Control.

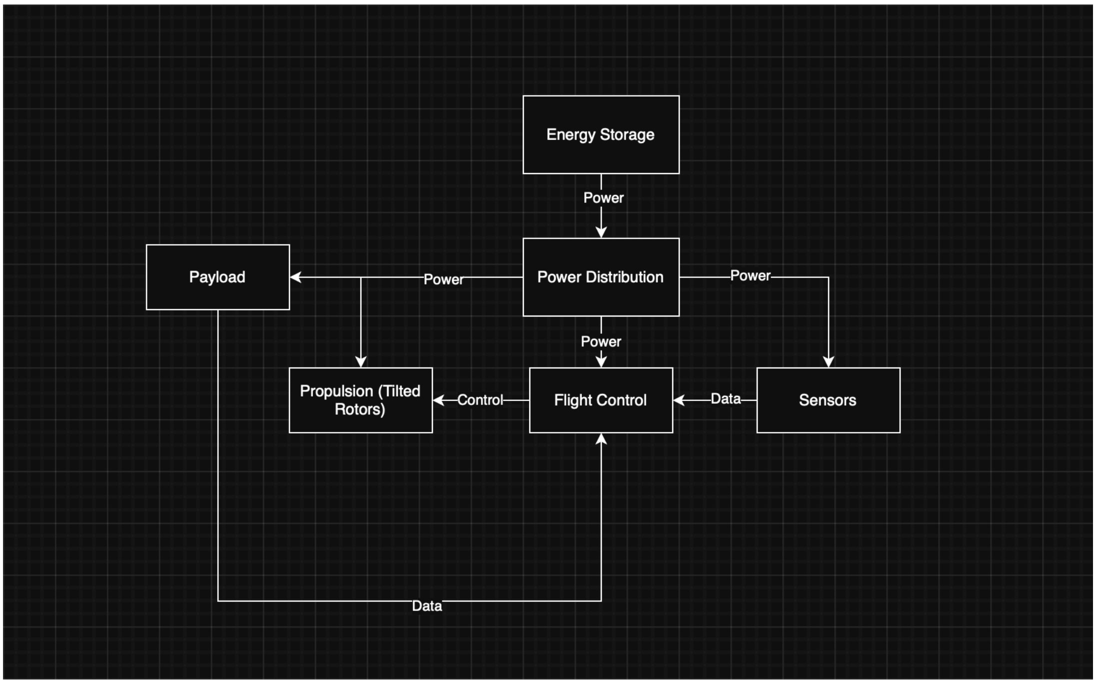
This separation highlights a key characteristic of the architecture: hover and cruise are performed by different hardware. While this simplifies VTOL operation and can improve hover efficiency, it introduces inactive mass during cruise.




## 4.3 Tilt-Rotor Architecture

The tilt-rotor configuration uses a single integrated **Propulsion (Tilted Rotors)** subsystem for both vertical and forward flight. Separate lift and cruise propulsion blocks are therefore replaced by one shared propulsion block.

This architecture reduces inactive mass during cruise by reusing propulsion hardware, at the cost of increased mechanical and control complexity, which is reflected by the stronger coupling between propulsion and flight control.



## 4.4 Architectural Comparison

By keeping all non-propulsion subsystems identical, the block diagrams isolate the propulsion system as the primary architectural difference. This provides a clear basis for comparing energy flow, control complexity, and system-level trade-offs in the following analysis sections.

# 5. Hover Analysis

## 5.1 Assumptions

$$m = 5{,}5\,\text{kg}, \quad g = 9{,}81\,\text{m.s}^2, \quad \rho = 1{,}225\,\text{kg.m}^3$$

Steady hover, zero climb rate, no wind, no ground effect. Equal thrust sharing between rotors. Induced power estimated using actuator disk theory.

## 5.2 Total Hover Thrust

In steady hover:

$$T_{total} = W = mg$$

$$T_{total} = 5{,}5 \cdot 9{,}81 = 54{,}0\,\text{N}$$

## 5.3 Quad-Lift + Pusher Configuration

Four dedicated lift rotors provide vertical thrust.

$$T_{rotor} = \frac{T_{total}}{4} = \frac{54{,}0}{4} = 13{,}5\,\text{N}$$

Assume lift rotor radius:

$$R = 0{,}15\,\text{m}$$

$$A_{rotor} = \pi R^2 = \pi (0{,}15)^2 = 0{,}0707\,\text{m}^2$$

Disk loading:

$$DL = \frac{T_{rotor}}{A_{rotor}} = \frac{13.5}{0{,}0707} \approx 191\,\text{N.m}^2$$

Induced hover power per rotor:

$$P_{hover,\,rotor} = \frac{T_{rotor}^{\frac{3}{2}}}{\sqrt{2\rho A_{rotor}}} = \frac{(13{,}5)^{\frac{3}{2}}}{\sqrt{2 \cdot 1{,}225 \cdot 0{,}0707}} \approx 119\,\text{W}$$

**Total hover power:**

$$P_{hover,\,total} = 4 \cdot 119 \approx 476\,\text{W}$$

## 5.4 Tilt-Rotor Configuration

Two tilt rotors provide vertical thrust during hover.

$$T_{rotor} = \frac{T_{total}}{2} = \frac{54{,}0}{2} = 27{,}0\,\text{N}$$

Assume smaller rotor radius due to cruise optimization:

$$R = 0{,}12\,\text{m}$$

$$A_{rotor} = \pi (0{,}12)^2 = 0{,}0452\,\text{m}^2$$

Disk loading:

$$DL = \frac{27{,}0}{0{,}0452} \approx 597\,\text{N.m}^2$$

Induced hover power per rotor:

$$P_{hover,\,rotor} = \frac{(27{,}0)^{\frac{3}{2}}}{\sqrt{2 \cdot 1{,}225 \cdot 0{,}0452}} \approx 421\,W$$

**Total hover power:**

$$P_{hover,\,total} = 2 \cdot 421 = 842\,W$$

### 5.5 Hover Power Comparison

$$\boxed{\begin{aligned} P_{hover,\,quad\ lift} &\approx 476\ W \\ P_{hover,\,tilt\ rotor} &\approx 842\ W \end{aligned}}$$

The quad-lift + pusher architecture exhibits significantly lower hover power due to lower disk loading enabled by multiple large-diameter lift rotors. The tilt-rotor configuration incurs a hover power penalty driven by higher thrust per rotor and reduced disk area.

## 6. Cruise Analysis

$$g = 9{,}81\ m.s^{2} \ ,\ \rho = 1{,}225\ kg.m^{3}$$

### 6.1 Assumptions

Steady, level cruise flight. Lift generated by the fixed wing. Shaft power estimated; propulsion losses neglected.

Cruise speed:

$$V = 20\,m.s$$

Wing reference area:

$$S = 0{,}40\,m^{2}$$

Aerodynamic coefficients:

$$C_{L,\,cruise} = 0{,}5 \ ,\ C_{D_0,\,tilt} = 0{,}04 \ ,\ k = 0{,}08$$

### 6.2 Quad-Lift + Pusher Configuration (Cruise Penalties)

Inactive lift propulsion mass:

$$\Delta m_{lift} = 0{,}4\ kg$$

Total mass:

$$m_{quad} = 5{,}9\ kg$$

Weight:

$$W_{quad} = 57{,}9\ N$$

Increased parasitic drag due to lift rotors and arms:

$$C_{D0,quad} = 0{,}05$$

Drag coefficient:

$$C_D = C_{D0,quad} + kC_L^2$$
$$C_D = 0{,}05 + 0{,}08 \cdot 0{,}25 = 0{,}07$$

Drag force:

$$D_{quad} = \frac{1}{2}\rho V^2 S C_D$$
$$D_{quad} = \frac{1}{2} \cdot 1{,}225 \cdot 20^2 \cdot 0{,}40 \cdot 0{,}07 \approx 6{,}9\ N$$

**Cruise power:**

$$P_{cruise,quad} = D_{quad} \cdot N$$
$$P_{cruise,quad} = 138\ W$$

## 6.3 Tilt-Rotor Configuration (Baseline Cruise)

Total mass:

$$m_{tilt} = 5{,}5\ kg$$

Weight:

$$W_{tilt} = mg = 54{,}0\ N$$

Lift condition:

$$L = \frac{1}{2}\rho V^2 S C_L$$
$$L = \frac{1}{2} \cdot 1{,}225 \cdot 20^2 \cdot 0{,}40 \cdot 0{,}5 = 49\ N$$

Drag coefficient:

$$C_D = C_{D0,tilt} + kC_L^2$$
$$C_D = 0{,}04 + 0{,}08 \cdot 0{,}25 = 0{,}06$$

Drag force:

$$D_{tilt} = \frac{1}{2}\rho V^2 S C_D$$
$$D_{tilt} = \frac{1}{2} \cdot 1{,}225 \cdot 20^2 \cdot 0{,}40 \cdot 0{,}06 \approx 5{,}9\ N$$

**Cruise power:**

$$P_{cruise,tilt} = D_{tilt} \cdot V$$
$$P_{cruise,tilt} \approx 118\ W$$

### 6.4 Cruise Performance Comparison

$$P_{cruise,quad} \approx 138\,W$$
$$P_{cruise,tilt} \approx 118\,W$$

The quad-lift + pusher configuration exhibits higher cruise power due to inactive lift propulsion mass and increased parasitic drag. The tilt-rotor configuration benefits from propulsion hardware reuse and a cleaner aerodynamic profile, resulting in improved cruise efficiency and endurance potential.

# 7. Trade-Off Matrix

| Criterion | Quad-Lift + Pusher | Tilt-rotor |
|---|---|---|
| Hover Power | ≈ 476 W | ≈ 842 W |
| Cruise power | ≈ 138 W | ≈ 118 W |
| VTOL Performance | Better | Worse |
| Cruise efficiency | Worse | Better |
| Mechanical Complexity | Lower | Higher |
| Recommended mission fit | VTOL-heavy ISR | Cruise-heavy ISR |

# 8. Conclusion

This project presented a system-level trade study of two VTOL propulsion architectures for a small fixed-wing ISR UAV: a quad-lift + pusher configuration and a tilt-rotor configuration. The comparison was based on simplified but physics-based hover and cruise performance estimates under a common mission definition.

The hover analysis showed that the quad-lift + pusher architecture requires substantially lower power during vertical flight due to lower disk loading enabled by multiple large-diameter lift rotors. The tilt-rotor configuration exhibits a higher hover power requirement, as fewer and smaller rotors must generate a larger share of the total thrust. Since VTOL operation represents the most power-critical flight phase, this difference strongly influences propulsion sizing and operational margins.

In cruise, both architectures require significantly less power than in hover. However, the quad-lift + pusher configuration incurs a moderate cruise power penalty due to inactive lift propulsion mass and increased parasitic drag. The tilt-rotor configuration benefits from propulsion hardware reuse and a cleaner aerodynamic configuration, resulting in improved cruise efficiency.

Based on the quantitative results and qualitative considerations such as mechanical complexity and robustness, the quad-lift + pusher architecture is assessed as the more suitable baseline solution for the defined ISR mission. The tilt-rotor configuration may be preferable for missions with a stronger emphasis on long-endurance cruise performance.

## 9. References and Image Credits

All figures are used for conceptual illustration only and do not represent the analyzed design.


1. Introduction to Aerospace Flight Vehicles – *Vertical Takeoff and Landing (VTOL) Aircraft*
   Embry-Riddle Aeronautical University
   https://eaglepubs.erau.edu/introductiontoaerospaceflightvehicles/
2. Austin, R. *Unmanned Aircraft Systems: UAVS Design, Development and Deployment*
   Wiley, 2010
3. Anderson, J. D. *Introduction to Flight*
   McGraw-Hill Education
4. Design and Analysis of Fixed-Wing VTOL UAV
   University Technical Report
   https://people.utm.my/mnazri/
5. Lift-to-Drag Ratio
   Wikipedia
   https://en.wikipedia.org/wiki/Lift-to-drag_ratio
6. Tilt-Rotor VTOL UAV Overview
   MDPI Aerospace
   https://www.mdpi.com/journal/aerospace
7. Example UAV configuration images used for conceptual illustration only, sourced from publicly available manufacturer and research material.

Figure 1: https://quantum-systems.com/vector-ai/
Figure 2: https://www.reuters.com/news/picture/war-from-above-aerial-views-of-russias-i-idJPRTS8PQPT/
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